

UNIVERSITÄT
LEIPZIG

Tutorial 1

Basic Programming Paradigms

Leipzig, 22.10.2026
Enny May

Outline

- General information
- Introduction to Shells
- Programming
- Questions

General information

- Content of the tutorial
  - Repetition of the content of the lecture
  - Programming
  - Code tracing
  - Exam preparation
- Every Thursday 9:15 in SR 7
- Questions to: e.may@studserv.uni-leipzig.de

General information

- Questions via Particify:



particify.uni-leipzig.de

9261 7839



Introduction to Shells

What are shells?

- Human-machine interface
- Allows communication with the Kernel
- Acts as command line interpreter
- Runs on top of the Kernel
- Many different shell-programs
  - Bash
  - C-Shell
  - …
- *echo $0* ~ shows active shell
- env („environment“) ~ provides information about environment variables
- *man <command>* („manual“) ~ displays more information about the specified command

Quelle: Graiger, Christian: Bash Programmierung Einstieg und professioneller Einsatz, entwickler.press, 2009.

Navigation

- *ls* („list“) ~ lists the content of the current directory
- *pwd* („print working directory“) ~ prints the absolute path of the current working directory
- *cd* („change directory“) ~ changes to the user’s home directory
  - *cd <path>* ~ follows specified path to get to the directory
  - *cd ..* ~ moves up one directory level to the parent directory

Creating and deleting files and folders

- *mkdir <name>* („make directory“) ~ creates a directory with the specified name
- *rmdir <name>* („remove directory“) ~ deletes an empty directory
- *touch <name>* ~ creates a new file, if the file does not already exist
- *rm <name>* („remove“) ~ deletes a file
  - *rm -r <directory name>* („remove recursively“) ~ removes directories and their content recursively

Copying, moving and renaming files and folders

- *cp <name> <path>* („copy“) ~ copies file or directory into the directory, which is described by the path
- *mv <name> <new name>* („move“) ~ renames file or folder
- *mv <name> <path>* („move“) ~ moves file or directory into the directory, which is described by the path

Editing files

- *lite-xl <name>* ~ opens the specified file in lite-xl
  - recommendation: *lite-xl <name> &* ~ opens editor as a background process
- alternatively: *nano <name>* ~ opens the specified file in nano
- alternatively: *vim <name>* ~ opens the specified file in vim
  - Exiting vim:
    - Esc, type :q and press enter
  - Safe changes:
    - Esc, type :w and press enter

Compiling and running programs

- *g++ <file name>* ~ compiles a file
  - *g++ <file name> -o <executable name>* ~ compiles file and names the resulting executable
  - Without -o flag: name of the executable is *a.out*
- *./<executable name>* ~ runs executable
  - Without -o Flag: *./a.out* ~ runs the executable

Input und Output Redirection

Output Redirection

- *./a.out > text.txt* ~ redirects output of the standard output streams to text.txt
- *./a.out 2> text.txt* ~ redirects the output of the error streams to text.txt
- *./a.out &> text.txt* ~ redirects the output of both streams to text.txt

Input Redirection

- *./a.out < text.txt* ~ the program a.out receives input from text.txt

Exercise (3 Minuten)

- Navigate to the BPP directory in your virtual machine
- Create a folder and name it „Tutorial“
- Navigate into this folder
- Create a file „test.cpp“
- Rename this file to „HelloWorld.cpp“
- Open „HelloWorld.cpp“ in an editor of your choice

Exercise - solution

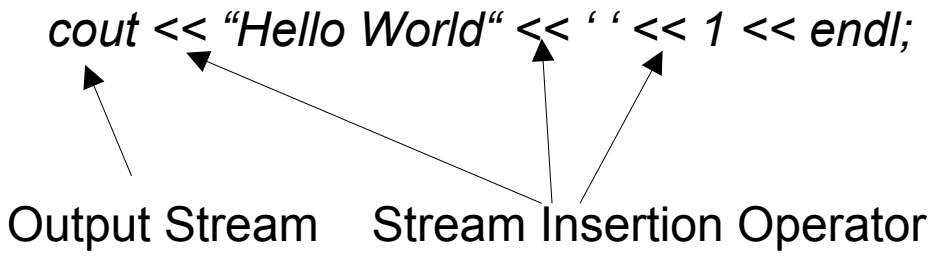
- Navigate to the BPP directory in your virtual machine
  - *cd ~/BPP*
- Create a folder and name it „Tutorial“
  - *mkdir Tutorial*
- Navigate into this folder
  - *cd Tutorial*
- Create a file „test.cpp“
  - *touch test.cpp*
- Rename this file to „HelloWorld.cpp“
  - *mv test.cpp HelloWorld.cpp*
- Open „HelloWorld.cpp“ in an editor of your choice
  - z.B.: *lite-xl HelloWorld.cpp*

Programming

Exercise 1 – Hello World

- Write a program that outputs „Hello World“ into the Terminal.

Standard Output Stream



- Stream insertion operators are used between literals of different data types, variables and stream manipulators
- *cout* sends a stream of characters to the terminal

Exercise 2 – Standard Output (5 minutes)

- Write a program that outputs your name, your student number and course of study.
- Create a variable that stores your first and last name and use it for outputting your name.

Notes

- Use following include directive in order to use strings: *#include <string>*

Exercise 3 – Countdown (5 minutes)

- Write a C++ program, which counts down from any positive integer to zero.

Notes

- Declare, define and initialize an integer:
  - int number {42};
- Syntax of while-loops:
  - while(condition) { //do something }

Questions?



UNIVERSITÄT
LEIPZIG

THANK YOU!

Enny May

Professur für Wirtschaftsinformatik, insbesondere Softwareentwicklung für Wirtschaft und Verwaltung

e.may@studserv.uni-leipzig.de